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Welcome to the bash simulator. Type a command and press Enter.
- Challenge 1: Rung 1 - check yourself first: confirm this workstation still has a valid address. - (fresh filesystem for this challenge)
student@ivy-workstation:/home/student$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 02:1a:7c:44:0b:5e brd ff:ff:ff:ff:ff:ff
    inet 10.20.5.42/24 brd 10.20.5.255 scope global eth0
        valid_lft forever preferred_lft forever
✓ Your own address is fine: 10.20.5.42/24. Rung 1 clear - the problem is not your machine.
→ Press Next (or Enter on an empty line) to continue to the next challenge.
- Challenge 2: Still rung 1 - confirm you know where your traffic leaves from. - (fresh filesystem for this challenge)
student@ivy-workstation:/home/student$ ip route
default via 10.20.5.1 dev eth0
10.20.5.0/24 dev eth0 proto kernel scope link src 10.20.5.42
✓ Gateway is 10.20.5.1, exactly as in Lab 01. Nothing has changed on your side.
→ Press Next (or Enter on an empty line) to continue to the next challenge.
- Challenge 3: Rung 2 - can you get out at all? Ping your gateway before you blame anything further away. - (fresh filesystem for this challenge)
student@ivy-workstation:/home/student$ ping 10.20.5.1
PING 10.20.5.1 (10.20.5.1) 56(84) bytes of data.
64 bytes from grid-gateway (10.20.5.1): icmp_seq=1 ttl=64 time=1.000 ms
64 bytes from grid-gateway (10.20.5.1): icmp_seq=2 ttl=64 time=1.200 ms
student@ivy-workstation:/home/student$

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student@ivy-workstation:/home/student$ ping 10.20.5.1
PING 10.20.5.1 (10.20.5.1) 56(84) bytes of data.
64 bytes from grid-gateway (10.20.5.1): icmp_seq=1 ttl=64 time=1.000 ms
64 bytes from grid-gateway (10.20.5.1): icmp_seq=2 ttl=64 time=1.200 ms
64 bytes from grid-gateway (10.20.5.1): icmp_seq=3 ttl=64 time=1.100 ms
64 bytes from grid-gateway (10.20.5.1): icmp_seq=4 ttl=64 time=1.300 ms

--- 10.20.5.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.000/1.150/1.300/0.100 ms
✓ The gateway answers. Rungs 1 and 2 are clear, so whatever is wrong is past your own doorstep.
→ Press Next (or Enter on an empty line) to continue to the next challenge.
- Challenge 4: Confirm the report: try to ping relay-station.grid.local. - (fresh filesystem for this challenge)
student@ivy-workstation:/home/student$ ping relay-station.grid.local
PING relay-station.grid.local (10.20.5.30) 56(84) bytes of data.

--- relay-station.grid.local ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3005ms
✓ Look closely: the name resolved to 10.20.5.30 and then every packet was lost. Two different things.
→ Press Next (or Enter on an empty line) to continue to the next challenge.
- Challenge 5: Rule DNS in or out: look up relay-station.grid.local directly. - (fresh filesystem for this challenge)
student@ivy-workstation:/home/student$ dig relay-station.grid.local
; <<>> DiG 9.18.24 <<>> relay-station.grid.local
;; global options: +cmd
;; Got answer:
;; HEADER: opcode=QUERY status=NOERROR id=41207
student@ivy-workstation:/home/student$

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Challenge 5: Make DNS an or even look up relay-station.grid.local directly. (fresh filesystem for this challenge)
student@ivy-workstation:/home/student$ dig relay-station.grid.local
; <<> DiG 9.18.24 <<> relay-station.grid.local
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 41207
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
relay-station.grid.local.      IN A

;; ANSWER SECTION:
relay-station.grid.local.  3600 IN A 10.20.5.30

;; Query time: 1 msec
;; SERVER: 10.20.5.10#53(10.20.5.10)
✓ NOERROR with an A record. DNS is doing its job perfectly. The name is not the problem.
→ Press Next (or Enter on an empty line) to continue to the next challenge.
- Challenge 6: Skip the name entirely – ping the address 10.20.5.30. – (fresh filesystem for this challenge)
student@ivy-workstation:/home/student$ ping 10.20.5.30
PING 10.20.5.30 (10.20.5.30) 56(84) bytes of data.

--- 10.20.5.30 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3005ms
✓ Still 100% loss with no name involved. That proves the fault is the machine, not the lookup.
→ Press Next (or Enter on an empty line) to continue to the next challenge.
student@ivy-workstation:/home/student$
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```
student@ivy-workstation:/home/student$ ping 10.20.5.30
PING 10.20.5.30 (10.20.5.30) 56(84) bytes of data.

--- 10.20.5.30 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3005ms
✓ Still 100% loss with no name involved. That proves the fault is the machine, not the lookup.
→ Press Next (or Enter on an empty line) to continue to the next challenge.
– Challenge 7: Find out how far your packets actually get. – (fresh filesystem for this challenge)
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```
student@ivy-workstation:/home/student$ traceroute relay-station.grid.local
traceroute to relay-station.grid.local (10.20.5.30), 8 hops max, 60 byte packets
```

```
 1 grid-gateway (10.20.5.1)  1.000 ms  1.200 ms  1.100 ms
 2 * * *
 3 * * *
 4 * * *
 5 * * *
 6 * * *
 7 * * *
 8 * * *
```

✓ Hop 1 answers, then stars. Your packets leave fine – nothing comes back from the far end.

→ Press Next (or Enter on an empty line) to continue to the next challenge.

– Challenge 8: Trace to a host you have no reason to suspect – cloud-heights.grid.local – as your control. – (fresh filesystem for this challenge)

```
student@ivy-workstation:/home/student$ traceroute cloud-heights.grid.local
traceroute to cloud-heights.grid.local (10.20.7.80), 8 hops max, 60 byte packets
```

```
 1 grid-gateway (10.20.5.1)  1.000 ms  1.200 ms  1.100 ms
 2 grid-core (10.20.0.1)    4.000 ms  4.200 ms  4.100 ms
```

```
student@ivy-workstation:/home/student$
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```
traceroute to cloud-heights.grid.local (10.20.7.80), 8 hops max, 60 byte packets
```

```
 1 grid-gateway (10.20.5.1)  1.000 ms  1.200 ms  1.100 ms
 2 grid-core (10.20.0.1)    4.000 ms  4.200 ms  4.100 ms
 3 cloud-heights.grid.local (10.20.7.80) 12.000 ms 12.200 ms 12.100 ms
```

✓ This one completes in three hops. Same first hop as the failed trace, so the local path out is healthy – the fault is at the far end.

→ Press Next (or Enter on an empty line) to continue to the next challenge.

– Challenge 9: Compare with the decommissioned host: ping relay-station-old.grid.local. – (fresh filesystem for this challenge)

```
student@ivy-workstation:/home/student$ ping relay-station-old.grid.local
```

```
ping: relay-station-old.grid.local: Name or service not known
```

✓ A completely different failure: Name or service not known. Nothing was ever sent, because there was no address to send to.

→ Press Next (or Enter on an empty line) to continue to the next challenge.

– Challenge 10: Prove the rest of the network is healthy before you escalate. – (fresh filesystem for this challenge)

```
student@ivy-workstation:/home/student$ ping foundry-archive.grid.local
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PING foundry-archive.grid.local (10.20.5.20) 56(84) bytes of data.
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64 bytes from foundry-archive.grid.local (10.20.5.20): icmp_seq=1 ttl=64 time=2.000 ms
64 bytes from foundry-archive.grid.local (10.20.5.20): icmp_seq=2 ttl=64 time=2.200 ms
64 bytes from foundry-archive.grid.local (10.20.5.20): icmp_seq=3 ttl=64 time=2.100 ms
64 bytes from foundry-archive.grid.local (10.20.5.20): icmp_seq=4 ttl=64 time=2.300 ms
```

```
--- foundry-archive.grid.local ping statistics ---
```

```
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
```

```
rtt min/avg/max/mdev = 2.000/2.150/2.300/0.100 ms
```

✓ Your control test replies. One host is down; the Grid is not.

```
student@ivy-workstation:/home/student$
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